Five exam-style calculation exercises

1. Discounted Return, MC Target, and TD(0) Update

An agent experiences the following rewards from time step $t = 0$:

$$R_1 = 2, R_2 = -1, R_3 = 4$$

After that, the episode terminates. Let $\gamma = 0.9$. Furthermore, assume

$$V(S_0) = 1.5, V(S_1) = 3.0, \alpha = 0.1.$$

First compute the Monte Carlo return G_0, then the TD(0) error $\delta_0$for the transition $S_0 \to S_1$, and finally the updated value $V(S_0)$.

Short solution

$$G_0 = 2 + 0.9(-1) + 0.9^2 \cdot 4 = 2 - 0.9 + 3.24 = 4.34.$$

The TD(0) error is

$$\delta_0 = R_1 + \gamma V(S_1) - V(S_0) = 2 + 0.9 \cdot 3.0 - 1.5 = 3.2.$$

Therefore,

$$V_{\text{new}}(S_0) = 1.5 + 0.1 \cdot 3.2 = 1.82.$$

This tests the difference between Monte Carlo methods, which use the full return, and TD methods, which bootstrap.

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2. Bellman Optimality Backup / Value Iteration

Consider a state $s$with two actions $a_1$and a_2. Let $\gamma = 0.9$. The current values of the successor states are

$$V_k(x) = 4, V_k(y) = 0.$$

For action a_1:

$$r(s, a_1) = 1, P(x \mid s, a_1) = 0.7, P(y \mid s, a_1) = 0.3.$$

For action a_2:

$$r(s, a_2) = 0, P(x \mid s, a_2) = 0.5, P(y \mid s, a_2) = 0.5.$$

Compute $V_{k+1}(s)$using a Bellman optimality backup and determine the greedy action.

Short solution

For a_1:

$$1 + 0.9(0.7 \cdot 4 + 0.3 \cdot 0) = 1 + 0.9 \cdot 2.8 = 3.52.$$

For a_2:

$$0 + 0.9(0.5 \cdot 4 + 0.5 \cdot 0) = 1.8.$$

Thus,

$$V_{k+1}(s) = \max\,(3.52,1.8) = 3.52.$$

The greedy action is a_1.

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3. SARSA vs. Q-Learning

An agent is in state S_t, selects action A_t, receives reward

$$R_{t+1} = 1$$

and transitions to S_{t+1}. Let

$$Q(S_t, A_t) = 2, \alpha = 0.5, \gamma = 0.9.$$

In the next state, there are two possible actions:

$$Q(S_{t+1}, \text{left}) = 5, Q(S_{t+1}, \text{right}) = 2.$$

The next action actually selected under the behavior policy is right. Compute the updated $Q(S_t, A_t)$once using SARSA and once using Q-Learning.

Short solution

SARSA uses the action actually selected next:

$$y_{\text{SARSA}} = 1 + 0.9 \cdot Q(S_{t+1}, \text{right}) = 1 + 0.9 \cdot 2 = 2.8.$$
$$Q_{\text{new}} = 2 + 0.5(2.8 - 2) = 2.4.$$

Q-Learning instead uses the greedy action in the next state:

$$y_{\mathrm{Q}} = 1 + 0.9 \cdot \max(5,2) = 1 + 4.5 = 5.5.$$
$$Q_{\mathrm{new}} = 2 + 0.5(5.5 - 2) = 3.75.$$

This is a very typical calculation exercise because it directly shows the difference between on-policy and off-policy learning.

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4. DQN Target vs. Double DQN Target

For a non-terminal transition, assume:

$$r = 1, \gamma = 0.99, Q_{\mathrm{current}}(s,a) = 4.$$

For the next state s', the online network outputs:

$$Q_{\mathrm{online}}(s',\cdot) = [3,5,4],$$

and the target network outputs:

$$Q_{\mathrm{target}}(s',\cdot) = [6,2,4].$$

Compute the standard DQN target, the Double DQN target, and the squared error loss $(y - Q_{\mathrm{current}})^2$for both.

Short solution

In standard DQN, we take the maximum over the target network:

$$\max\ Q_{\mathrm{target}}(s',\cdot) = 6.$$

Therefore,

$$y_{\mathrm{DQN}} = 1 + 0.99 \cdot 6 = 6.94.$$
$$L_{\mathrm{DQN}} = (6.94 - 4)^2 = 2.94^2 = 8.6436.$$

In Double DQN, the online network selects the action:

$$\arg\ \max\ Q_{\mathrm{online}}(s',\cdot) = 2$$

assuming we index actions as the second action, i.e. the value 5. However, this selected action is evaluated using the target network:

$$Q_{\mathrm{target}}(s', a_2) = 2.$$

Thus,

$$\begin{aligned} y_{\text{DoubleDQN}} &= 1 + 0.99 \cdot 2 = 2.98. \\ L_{\text{DoubleDQN}} &= (2.98 - 4)^2 = (-1.02)^2 = 1.0404. \end{aligned}$$

This example tests whether you understand that Double DQN separates **action selection** from **action evaluation**.

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5. GAE and PPO Clipping

Given the TD errors

$$\delta_0 = 1, \delta_1 = -0.5, \delta_2 = 2.$$

Let

$$\gamma = 0.9, \lambda = 0.8.$$

Compute the Generalized Advantage Estimate

$$\hat{A}_0^{GAE} = \delta_0 + \gamma\lambda\delta_1 + (\gamma\lambda)^2\delta_2.$$

Then assume for PPO:

$$\hat{A}_0 = \hat{A}_0^{GAE}, r_0(\theta) = 1.3, \epsilon = 0.2.$$

Compute the PPO clipped surrogate term

$$\min\left(r_0(\theta)\hat{A}_0, \text{clip}(r_0(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_0\right).$$

Short solution

First,

$$\gamma\lambda = 0.9 \cdot 0.8 = 0.72.$$

Therefore,

$$\begin{aligned} \hat{A}_0^{GAE} &= 1 + 0.72(-0.5) + 0.72^2 \cdot 2. \\ &= 1 - 0.36 + 0.5184 \cdot 2 = 1 - 0.36 + 1.0368 = 1.6768. \end{aligned}$$

For PPO, $r_0(\theta) = 1.3$is clipped to the interval

$$[1-\epsilon, 1+\epsilon] = [0.8, 1.2].$$

So,

$$\text{clip}(1.3, 0.8, 1.2) = 1.2.$$

Unclipped:

$$1.3 \cdot 1.6768 = 2.17984.$$

Clipped:

$$1.2 \cdot 1.6768 = 2.01216.$$

Since the advantage is positive, PPO takes the smaller value:

$$L^{CLIP} = 2.01216.$$

This is very exam-style because it combines TD errors, advantage estimation, and the trust-region-like restriction used in PPO.